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Adherence to the mediterranean diet in adults: the role of environmental and sustainable food literacy

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Abstract

Aim This study aimed to evaluate the relationship between environmental literacy, sustainable food literacy, and adherence to the Mediterranean diet (MD) in adults.

Method Data were collected via a web-based questionnaire created using Google Forms between April and August 2024 in Erzurum, Turkey. A total of 394 adults participated through snowball sampling. The questionnaire included sections on demographic information, anthropometric measurements, the Environmental Literacy Scale for Adults (ELSA), the Sustainable Food Literacy Scale (SFLS), and the Mediterranean Diet Adherence Screener (MEDAS).

Results 38.6% of the participants were found to adhere to the Mediterranean Diet (MD). Married participants were more likely to have adhered to the MD than single ones (odds ratio [OR]:0.41, $p=0.006$). Adherence to the MD was associated with higher education duration (OR = 1.05, $p=0.011$), a higher SFLS total score (OR = 1.01, $p=0.025$), and a higher ELSA score (OR = 1.04, $p=0.004$).

Conclusion The MD is a sustainable dietary model that enhances health and is environmentally friendly. For this reason, it may be possible to direct society towards healthier and less environmentally damaging food choices through training to increase individuals' sustainable literacy and environmental literacy levels.

Keywords Mediterranean diet, Environmental literacy, Sustainable food literacy, Diet quality, Adults

Introduction

Climate change and environmental pollution are complex issues with many facets. Environmental education is the most effective approach to mitigating environmental challenges, which can help raise environmentally conscious individuals [1]. According to Roth (1992), an environmentally literate individual is someone who can understand the relationship between ecological and societal systems, believes in the integrity of human and nature, is conscious of the environmental implications of technological advancements and knows that he/she can learn about the environment throughout his/her life.

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Knowledge, attitudes, behaviours, and awareness are the four core components of environmental literacy [2].

Our planet is suffering from the rapid growth of the world's population, increased food consumption, and changing dietary habits. This underscores the urgent need to promote food systems that are not only nutritious but also sustainable. Accordingly, the world should adopt an adequate, balanced, nutritious, sustainable diet that enhances food security, promotes local or seasonal agricultural production, and exhibits a minimal environmental footprint [3]. One of the most widely recognized dietary patterns that aligns with these principles is the Mediterranean dietary pattern, recognized for its numerous health advantages and its low environmental impact [4]. The Mediterranean diet (MD) focuses on the intake of natural and domestic species, with seasonal, local goods and traditional cooking methods significantly contributing to biodiversity preservation and environmental sustainability [5].

The Food and Agriculture Organisation (FAO) stated that to achieve sustainable nutrition, consumer behaviour must change and strategies that positively impact human health and ecosystems should be promoted to facilitate the transition from existing food systems [6]. Similarly, the World Health Organization's 2030 Sustainability goals include improving food security and enhancing nutrition, providing sustainable food production systems, protecting the environment, and taking measures against the climate crisis [7]. In this context, it is essential to create food literacy initiatives that will provide people with the knowledge and skills they need to understand food systems, make informed dietary choices, and develop healthy eating behaviours. By equipping individuals with these competencies, food literacy can improve personal health and well-being and enhance the sustainability of agricultural frameworks and the safeguarding of ecological systems [8].

Prospective and large sample studies have demonstrated that sustainable eating practices are linked to less negative environmental effects [9–11]. However, studies exploring the relationship between sustainable eating practices and food literacy remain limited. Current data indicates that food and nutrition literacy can provide the knowledge necessary to understand the factors dietary habits and their environmental consequences [12, 13]. Nevertheless, the literature review revealed that no research has examined the potential relationship between the degree of adherence to the MD and the level of environmental and food literacy. It is essential to understand this relationship since it may influence both public policy and academic studies. By investigating the connection between environmental literacy, sustainable food literacy, and adherence to the MD, this study aims to provide insights that can guide

future educational programs, policies, and research initiatives to promote environmental sustainability and healthy eating. Thus, the purpose of this study is to assess the relationship between adult adherence to the MD, environmental literacy, and sustainable food literacy.

Materials and methods

Study design and participants

This descriptive and cross-sectional study recruited 394 adults who consented to participate in Erzurum, Turkey. The data for this study were collected through a web-based questionnaire created using Google Forms. The online questionnaire was disseminated via various social media channels such as Twitter, Facebook, WhatsApp, and Instagram. Data collection was carried out between April and August 2024. A snowball sampling method was employed to recruit participants. Individuals who voluntarily agreed to participate by selecting the "I consent to participate in this study voluntarily" option at the beginning of the form and who completed the questionnaire in full were included in the final sample. Inclusion criteria for the study were: being between 18 and 65 years of age, not having any diagnosed chronic illnesses, psychological or eating disorders, not following a special diet, and providing informed consent to participate. Participants were excluded if they were pregnant or breastfeeding, did not meet the age criteria, or were adhering to a special dietary pattern. A total of 457 individuals initially accessed the online survey. However, based on the predefined exclusion criteria, 63 participants were excluded from the study: 30 due to incomplete responses, 15 due to chronic illnesses, 8 due to a history of psychological disorders, and 10 due to following a special diet. As a result, the final analysis was conducted with a sample of 394 participants.

The research was conducted with "Ethics Committee Approval" from the Erzurum Technical University Ethics Committee (Meeting Number: 03, Decision Number: 17, 2024). The study was carried out following the principles outlined in the Helsinki Declaration. Informed consent was obtained electronically at the beginning of the questionnaire.

Data collection

Several measures were used to evaluate the relationship between environmental literacy, sustainable food literacy, Mediterranean diet adherence, and variables such as age, gender, education, marital status, and BMI. The questionnaire form comprised a general information form, anthropometric measurements, dietary habits, the Environmental Literacy Scale for Adults (ELSA), the

Sustainable Food Literacy Scale (SFLS), and the Mediterranean Diet Adherence Screener (MEDAS).

Socio-demographic characteristics and anthropometric measurements

The socio-demographic characteristics, including age (in years), gender (male/female), years of education, and marital status (single or married) were questioned.

The participants self-reported their height (cm) and body weight (kg). The body mass index (BMI) was calculated by dividing body weight by the square of height and assessed according to the World Health Organization's BMI categorisation [14].

Environmental literacy scale for adults

The Environmental Literacy Scale for Adults (ELSA) was created by Atabek-Yiğit et al. (2014) to assess the environmental literacy levels of adults [1]. There are 20 items and three subscales on the scale: Environmental consciousness level (3, 10, 11, 12, 13, 15, 16, 17, 19), Environmental anxiety level (1, 2, 4, 5, 7, 8), and Environmental awareness level (6, 9, 14, 18, 20). Responses to the questions are provided on a Likert scale, with options ranging from "1" Strongly Agree, "2" Agree, "3" Undecided, "4" Disagree, to "5" Strongly Disagree. Items 4 and 16 are scored in the opposite direction.

All items are scored between 1 and 5, and the scale has a lowest possible score of 20 and a maximum score of 100. High scores on the test indicated high environmental literacy levels, whereas low scores indicated low environmental literacy levels in individuals. The scale's Cronbach's alpha was 0.881. The Cronbach's alpha values for the Environmental Consciousness, Anxiety, and Awareness subscales were 0.807, 0.765, and 0.715 [1].

Sustainable food literacy scale

The Sustainable Food Literacy Scale was developed by Teng and Chih (2022) with an emphasis on environmental sustainability and is intended to be used to assess people's tendencies to adopt sustainable diets [15]. Kubilay and Yüksel (2023) researched the validity, reliability, and adaptation of the scale in Turkish [16]. The scale has 26 items and a five-factor (Sustainable Food Knowledge 1, Sustainable Food Knowledge 2, Cooking and Kitchen Skills, Attitudes and Intentions for Action, and Action Strategies). Response options for the questions are measured on a Likert scale, ranging from ("1" strongly disagree, "2" disagree, "3" slightly disagree, "4" undecided, "5" slightly agree, "6" agree, and "7" strongly agree). There is no reverse item on the scale. The scores for each item range from 1 to 7. As the score increases, the level of sustainable food literacy increases. Cronbach's alpha value of the scale was determined as 0.941 [16].

Mediterranean diet adherence screener

Schröder et al. (2011) developed the Mediterranean Diet Adherence Scale (MEDAS) to measure adherence to the MD [17]. Pehlivanoglu et al. (2020) conducted the scale's validity, reliability, and Turkish adaptation [18]. There are 14 items on the scale. The scale is used to evaluate the type of oil used in meals, the daily intake of olive oil, portion sizes of fruits and vegetables, and the consumption of margarine, butter, and red meat. It also examines the weekly intake of wine, legumes, fish and seafood, nuts, cakes, and tomato sauce with olive oil, in addition to whether there is a preference for white meat over red meat. The final score is determined by taking 1 or 0 points for each question based on the amount of consumption. The range of MEDAS scores is 0 to 14, with 7 points indicating adherence to the MD. The Cronbach's alpha value for the scale was 0.829 [18].

Statistical analysis

SPSS version 23.0 software (IBM Corp., Armonk, NY, USA) was used to analyze the data. Kolmogorov-Smirnov tests, Skewness and Kurtosis, and visual analysis (histogram and probability graphs) were used to evaluate the data to see if they were normally distributed. While frequency (n) and percentage (%) were utilized to represent categorical variables, the mean ($\bar{X}$) and standard deviation (SD) were used for continuous variables. For comparisons between two independent groups, the Independent Samples t-test was used for normally distributed data, while the Mann-Whitney U test was employed for non-normally distributed data. In cases involving more than two groups, the Kruskal-Wallis H test was used. Categorical variables were analyzed using the Chi-square test. Pearson correlation analysis was used for the relationship between variables. A binary logistic regression analysis was conducted to evaluate the effect of factors on non-adherence vs. adherence to the MD and calculate the odds ratio (OR) and 95% confidence interval (CI). Non-adherence (code-0) vs. adherence to the MD (code-1) was used as the dependent variable, age (years), education duration (years), SFLS total score, ELSA score, gender (men-reference group vs. women), marital status (Single vs. Married-reference group), and BMI classification (underweight-reference group, normal weight, overweight/obese) were used as independent variables. The fit of the logistic regression model was evaluated with the Hosmer-Lemeshow test [19]. The threshold for statistical significance was determined at $p < 0.05$.

The G*Power (version 3.1.9.7, Universität Düsseldorf, Düsseldorf, Germany) was utilized for post-hoc power analysis following the study, employing the point

Table 1 General characteristics of the participants (n:394)

Variables	n	%
Gender		
Male	59	15.0
Female	335	85.0
Marital status		
Single	349	88.6
Married	45	11.4
Education status		
Secondary/High school	112	28.4
Bachelor/Master degree	282	71.6
BMI classification		
Underweight	39	9.9
Normal weight	265	67.3
Overweight/Obese	90	22.8
MEDAS classification		
No adherence (< 7)	242	61.4
Adherence (≥ 7)	152	38.6
	$\bar{x} \pm SS$	
Age (years)	23.4 ± 6.12	
Education duration (years)	14.4 ± 2.81	
BMI (kg/m ²)	22.8 ± 3.90	
SFLS total score	141.6 ± 19.62	
Sustainable food knowledge	48.1 ± 8.31	
Food skills	34.0 ± 5.45	
Attitudes	21.7 ± 3.91	
Action intent and action strategies	37.8 ± 6.41	
ELSA score	82.4 ± 8.35	
MEDAS score	6.1 ± 1.91	

BMI Body mass index, SFLS Sustainable food literacy scale, ELSA Enviromental literacy scale, MEDAS The Mediterranean Diet Adherence Scale

biserial correlation model, with a two-tailed effect size of $|\rho|=0.24$, α error probability of 0.05, and a determined power ($1-\beta$ error probability) of 99%.

Results

The general characteristics of the individuals are given in Table 1. Accordingly, 85.0% of the individuals were female, and 15.0% were male. The mean age was 23.4 ± 6.12 years, and the duration of education was 14.4 ± 2.81 years, 67.3% were in the normal BMI range, and 22.8% were overweight/obese. The mean MEDAS score of the individuals was 6.1 ± 1.91 , and 38.6% of them were found to adhere to the MD (≥ 7 scores). The Sustainable Food Literacy Scale total score was 141.6 ± 19.62 . The Environmental Literacy Scale score was 82.4 ± 8.35 (Table 1).

The results showed that women had a higher ELSA score (82.9 ± 7.73) than men (79.5 ± 10.87) ($p=0.025$). The bachelor/master degree group had a higher MEDAS score ($p=0.023$) than the secondary/high school group. According to educational status and BMI classification, there was no significant difference in the ELSA scores ($p>0.05$). However, a significant difference was observed

Table 2 The ELSA and MEDAS total scores according to gender, education and BMI classification

	ELSA score	p	MEDAS score	p
Gender				
Male	79.5 ± 10.87	0.025^Y	6.3 ± 1.75	0.293 ^Y
Female	82.9 ± 7.73		6.1 ± 1.94	
Education status ^Y				
Secondary/High school	82.4 ± 8.46	0.871 ^Y	5.9 ± 1.90	0.023^Y
Bachelor/Master degree	82.5 ± 8.14		6.4 ± 1.91	
BMI classification				
Underweight	81.0 ± 9.31	0.086*	5.6 ± 1.90	0.368*
Normal weight	82.1 ± 8.21		6.1 ± 1.92	
Overweight/Obese	84.0 ± 8.21		6.2 ± 1.89	

BMI Body mass index, ELSA Enviromental literacy scale, MEDAS The Mediterranean Diet Adherence Scale

^YIndependent sample t-test

* Kruskal Wallis H-test

The statistically significant p values are shown in **bold**

in the MEDAS scores based on educational status: participants with a Bachelor or Master degree had higher MEDAS scores compared to those with only secondary or high school education ($p=0.023$). No significant difference in MEDAS scores was found across BMI classifications ($p>0.05$) (Table 2).

Table 3 shows the distribution of the SFLS total score and its subcomponents by gender, education status, and BMI classification. The total SFLS score and subcomponents were statistically higher in women than men ($p<0.05$). However, no significant difference was found according to education status and BMI classification ($p>0.05$).

The factors associated with the participant's adherence to the MD are presented in Table 4. Accordingly, the results indicated that the mean age ($p=0.007$), total score of SFLS ($p=0.029$), the SFLS' subcomponent food skills ($p=0.001$), and the ELSA score ($p=0.005$) were significantly higher among participants adhering to the MD compared to those who were not.

The relationships among some variables, the SFLS, ELSA, and MEDAS, are shown in Table 5. SFLS total score was positively associated with all subcomponents of SFLS ($p<0.001$), ELSA ($p<0.001$), and MEDAS score ($p<0.05$). A positive relationship was also found between all subcomponents of SFLS and ELSA ($p<0.001$). MEDAS scores show a positive correlation with food skills ($p<0.001$) and action intent and action strategies ($p<0.05$), which are subcomponents of SFLS. Also, a significant positive relationship was observed between the ELSA score and MEDAS ($p<0.05$).

A binary logistic regression analysis was conducted to identify factors associated with adherence to the MD (Table 6). Age was positively associated with MD (OR = 1.05, 95% CI: 1.01–1.09, $p=0.005$). Similarly, education duration was positively associated with adherence

Table 3 The SFLS total and subcomponents scores according to gender, education status, and BMI classification

	SFLS total score	<i>p</i>	Sustain-able food knowledge	<i>p</i>	Food skills	<i>p</i>	Attitudes	<i>p</i>	Action intent and action strategies	<i>p</i>
Gender										
Male	131.2 ± 22.95	< 0.001^Y	45.3 ± 9.28	0.014^Y	30.5 ± 7.10	0.025[£]	20.4 ± 4.25	0.011^Y	34.9 ± 8.12	0.003^Y
Female	143.4 ± 18.41		48.6 ± 8.04		34.6 ± 4.86		21.9 ± 3.81		38.4 ± 5.93	
Education status										
Secondary/High school	142.7 ± 18.42	0.463 ^Y	48.6 ± 8.36	0.432 ^Y	34.4 ± 5.36	0.342 ^Y	21.7 ± 3.93	0.998 ^Y	38.0 ± 6.16	0.734 ^Y
Bachelor/Master degree	141.1 ± 20.09		47.9 ± 8.30		33.8 ± 5.48		21.7 ± 3.91		37.8 ± 6.51	
BMI classification										
Underweight	142.5 ± 22.45	0.698 [*]	49.2 ± 8.36	0.231 [*]	33.3 ± 5.52	0.161 [*]	22.2 ± 4.01	0.231 [*]	37.8 ± 7.12	0.792 [*]
Normal weight	141.5 ± 17.88		47.8 ± 7.82		33.9 ± 5.15		21.8 ± 3.68		38.0 ± 6.07	
Overweight/Obese	141.4 ± 23.17		48.5 ± 9.62		34.6 ± 6.21		21.0 ± 4.46		37.3 ± 7.08	

BMI: Body mass index, SFLS: Sustainable food literacy scale

^Y Independent Sample t-test[£] Mann-Whitney U test^{*} Kruskal Wallis H testThe statistically significant *p* values are shown in **bold****Table 4** Evaluation of some parameters according to individuals' adhere to the MD

	Mediterranean Diet Adherence Scale		
	No adher-ence to MD (< 7)	Adherence to MD (≥ 7)	<i>p</i>
Age (years)	22.6 ± 5.27	24.4 ± 7.15	0.007^Y
Gender (n,%)			
Male	37 (15.3)	22 (14.5)	0.892 ^a
Female	205 (84.7)	130 (85.5)	
Education duration (years)	14.1 ± 2.35	14.7 ± 3.40	0.069 ^Y
BMI (kg/m ²)	22.7 ± 3.59	23.0 ± 4.35	0.414 ^Y
SFLS total score	139.8 ± 18.17	144.4 ± 21.50	0.029^Y
Sustainable food knowledge	47.6 ± 7.87	48.8 ± 8.95	0.196 ^Y
Food skills	33.2 ± 5.25	35.2 ± 5.55	0.001^Y
Attitudes	21.5 ± 3.83	22.0 ± 4.02	0.149 ^Y
Action intent and action strategies	37.5 ± 6.25	38.4 ± 6.64	0.171 ^Y
ELSA total score	81.4 ± 7.80	84.0 ± 8.98	0.005^Y

BMI: Body mass index, SFLS: Sustainable food literacy scale, ELSA: Environmental literacy scale, MEDAS: The Mediterranean Diet Adherence Scale

^Y Independent Sample t-test^a Chi-square testThe statistically significant *p* values are shown in **bold**

to MD (OR = 1.05, 95% CI: 1.01–1.10, *p* = 0.011). Marital status was also significant, with married individuals demonstrating significantly greater adherence to the MD compared to single participants (OR = 0.41, 95% CI: 0.22–0.78, *p* = 0.006). Additionally, greater adherence was associated with a higher total SFLS score (OR = 1.01, 95% CI: 1.00–1.02, *p* = 0.025) and a higher ELSA score (OR = 1.04, 95% CI: 1.01–1.07, *p* = 0.004).

Discussion

The results of the current study suggest that participants' educational status is associated with adherence to the MD, and that Environmental Literacy and Sustainable Food Literacy scores are higher in women than in men. Additionally, differences in age, SFLS total score, and ELSA total score were observed between participants who adhered to the MD and those who did not. Upon further evaluation of factors influencing MD adherence, variables such as marital status, education duration, and ELSA and SFLS scores were identified significant variables.

In this study, women demonstrated significantly higher ELSA scores than men (Table 2). This finding is consistent with the prior investigation by Akçay and Şengül, who reported that female students had better environmental literacy levels than males [20]. Another study found no significant relationship between gender differences and ELSA total score. However, the score obtained from the environmental awareness subscale was found to be significantly higher in women [21]. These results suggest a potential gender-based difference in environmental consciousness, which might influence broader sustainable practices.

No significant association was found between educational status and ELSA scores in this study (Table 2). Similarly, a study conducted among students reported no statistically significant difference in environmental literacy among grade levels [20]. In contrast, another study found that participants who were members of environmental groups and had greater educational attainment also had higher levels of environmental awareness [21]. This result is consistent with previous studies that suggested women might become more involved in

Table 5 Correlation (r) of SFLS, ELSA, and MEDAS

	SFLS total score	Sustainable food knowledge	Food skills	Attitudes	Action intent and action strategies	ELSA score	MEDAS score	Age (year)	Education duration	BMI
SFLS total score	-									
Sustainable food knowledge	0.873*									
Food skills	0.773*	0.582*								
Attitudes	0.746*	0.536*	0.446*							
Action intent and action strategies	0.817*	0.555*	0.490*	0.599*						
ELSA score	0.533*	0.421*	0.474*	0.333*	0.478*					
MEDAS score	0.134 ^β	0.086	0.186*	0.049	0.112 ^β	0.160*				
Age (year)	-0.001	-0.014	0.104 ^β	-0.126 ^β	-0.021	0.133 ^β	0.165*			
Education duration	0.111 ^β	0.100 ^β	0.097	0.093	-0.071	0.058	0.158 ^β	0.109 ^β		
BMI	-0.073	-0.056	-0.015	-0.131 ^β	-0.059	0.052	0.052	0.363*	0.018	

SFLS: Sustainable food literacy scale, ELSA: Environmental literacy scale, MEDAS: The Mediterranean Diet Adherence Scale, BMI: body mass index

Pearson correlation coefficients

* $p < 0.001$, ^β $p < 0.05$ **Table 6** Logistic regression model for the odds of having non-adherence vs. adherence to the MD

		β	SE	OR (95% CI)	p
Age (year)		0.048	0.017	1.05 (1.01–1.09)	0.005
Gender	Male*				
	Female	0.064	0.292	1.07 (0.60–1.88)	0.825
Education duration (years)		0.053	0.021	1.05 (1.01–1.10)	0.011
Marital status	Single	-0.885	0.322	0.41 (0.22–0.78)	0.006
	Married*				
BMI classification	Underweight*				
	Normal weight	0.111	0.357	1.11 (0.55–2.24)	0.756
	Overweight/Obese	0.174	0.397	1.19 (0.55–2.59)	0.661
SFLS total score		0.012	0.005	1.01 (1.00–1.02)	0.025
ELSA score		0.038	0.013	1.04 (1.01–1.07)	0.004

*Reference group, Values in **bold** indicate significant at $p < 0.05$

household food procurement, improving their food literacy [22]. Similarly, in this study, both the SFLS total score and subcomponents were significantly higher among female participants, consistent with the findings of Kubilay et al. [16]. In one study, female participants reported consuming more foods classified as sustainable. However, researchers emphasize that factors such as cultural affiliations or taste limit the intake of some sustainable foods, such as tofu [23]. In our study findings, despite higher SFLS/ELSA in women, no gender difference was seen in MEDAS scores. This results may be due to the broader scope of sustainable diets (including Mediterranean, vegetarian, Nordic, plant-based, etc.) [24] compared to the Mediterranean diet model. Additionally, females tend to report stronger food knowledge and more positive attitudes towards sustainable diets; this apparent contradiction may indicate that these factors do not always translate into measurable dietary adherence in everyday life. Common home food surroundings, cultural influences, and social conventions may mitigate

gender-based disparities in real dietary behaviour. Unexpectedly, there was no significant difference in SFLS scores by educational attainment. Given that the majority of participants were young and university-educated, this could be because the sample was homogeneous. In these groups, media exposure or personal interests may have a greater impact on food literacy than formal education. These findings suggest that sustainable nutrition may be affected by multidimensional factors (environmental and individual) in addition to educational status.

The study conducted by Teng and Chih stated that sustainable food knowledge is significantly associated with education, although the strength of this relationship is very weak [15]. In line with this, sustainable food literacy levels were significantly lower among individuals with a high school education compared to those with university and postgraduate qualifications [16]. However, another study showed that education level did not significantly affect sustainable food literacy level [13]. Similarly, our study did not demonstrate a statistically significant

relationship between SFLS scores and educational levels (Table 3). The inconsistency in these findings may stem from differing study populations and the complexity of sustainable food literacy, which an array of socioeconomic and cultural determinants can shape.

Obesity is a sustainable development problem that is too comprehensive to be evaluated solely regarding health [25]. However, it appears that sustainable dietary models play a crucial role in mitigating the prevalence of obesity [26–28]. Our study found no significant association between SFLS scores and BMI values. This finding contrasts with the growing body of evidence suggesting that sustainable dietary models, such as the Mediterranean Diet (MD), can reduce the obesity risk [29]. One possible explanation for the absence of a significant association in our study is the limited number of participants classified as obese, which could have reduced the statistical power to detect such a relationship (Table 3).

MD is widely recognized as a sustainable diet due to its environmental benefits [4, 30]. One of its key features is the reduction in meat consumption, which studies have shown to have a less negative impact on the environment [31]. For instance, a study with over 6000 participants showed that participants who adhered more to the MD resulted in lower dietary CO₂ emissions than those who did not [32]. In another study, greater adherence to the MD is correlated with a decreased ecological footprint [33]. In a study by Rosi et al., a low correlation was observed between the students' compliance with the MD and their total carbon and ecological footprints [34]. In this study, consistent with the literature, it was found that environmental literacy and sustainable nutrition knowledge and attitudes increased depending on compliance with the MD.

Education status also emerged as a significant factor in adherence to the MD in our study. Participants with higher education levels exhibited greater adherence, consistent with other studies that have identified a positive correlation between education duration and MD adherence [35, 36]. In this study, education duration was among the factors affecting adherence to the MD (Tables 5 and 6). In addition, a significant difference was found between the MEDAS score and education levels of the participants categorized as Secondary School/High School and Bachelor/Master degree (Table 2). Higher education may facilitate greater awareness and implementation of dietary practices that promote health and sustainability.

Studies have shown that compliance with the MD is affected by age, gender, and marital status factors [37–41]. A research investigation involving university staff revealed a significant positive association between compliance with the MD and age [38]. In another study, compliance with the MD according to age was higher in the

elderly than in the young. Researchers attribute this to the fact that older individuals do not have a job, which facilitates cooking and eating at home easier [39]. A study conducted in Ireland with individuals over 50 years found a positive relationship between increasing age and female gender and compliance with the MD [40]. Similarly, in a cohort study found that compliance with the MD was lower among individuals younger 49 years and higher in individuals over 62 years [41]. In this study, age was among the factors affecting adherence to the MD, and a positive relationship was found between age and adherence to the MD (Tables 4, 5 and 6).

Numerous studies have demonstrated that adherence to MD is higher in women than men [36, 42, 43]. For instance, in a study of 1,333 adults, women had significantly higher MEDAS scores than men [42]. Similarly, research by Mieziene et al. in Lithuania found that women scored higher on MEDAS than men [36], and a study examining MD compliance during the COVID-19 pandemic also observed the same pattern [43]. However, a study conducted among university students found no significant gender difference in MEDAS scores [44]. In contrast to these studies, the present study revealed no statistically significant difference between males and females regarding MEDAS scores (Tables 4 and 6). The divergence may be attributed to the specific characteristics of our study population and the tools used to measure adherence, which may yield different outcomes depending on their sensitivity to gender differences.

Lastly, marital status was found to influence MD adherence in our study, with participants who lived alone exhibiting greater adherence than those in relationships. This result is consistent with research by Apostolaki et al., which also found a link between marital status and MD adherence [37]. However, another study did not show any relationship between age, gender, marital status and, MD adherence [45]. In a study, MD adherence was assessed using different tools in the same population. The influence of gender and age variables on adherence to the MD exhibited considerable variability. Therefore, the effect of the measurement tools used to determine the effect of factors such as gender, age, and marital status should be considered [46]. In the present study demonstrated that marital status was among the factors affecting compliance with the MD (Table 6).

Strengths and limitations

The research possesses several notable strengths. It represents one of the pioneering investigations evaluating the constructs of environmental literacy, sustainable food literacy, and adherence to the Mediterranean Diet (MD) among adults. Furthermore, the study identified significant associations between adherence to the MD and being married, having higher sustainable food literacy,

and higher environmental literacy. These results promote our knowledge of the potential relationships between sociocultural and demographic characteristics and sustainable and healthful eating habits. The research has several limitations: it employed a cross-sectional design, which implies that although it was unable to establish a causal relationship, it was nonetheless employed to evaluate the interrelations among the measured variables. Secondly, the self-reporting via online snowball sampling methodology was selected for data collection, which introduces the possibility of self-report biases and inaccuracies in reporting [47]. Thirdly, although we included male and female adults between 19 and 64 in the study, the sample may not be representative because it mainly consisted of younger adults and women. Finally, the individuals' self-reported data included measurements of the participants' height and body weight, which may have affected the accuracy of BMI.

Conclusion

In conclusion, this study identified significant associations between adherence to the Mediterranean Diet and factors such as longer education duration, higher sustainable food literacy, higher environmental literacy, and marital status. These findings suggest that certain sociodemographic and literacy-related factors may influence MD adherence in young adults. Nonetheless, the results highlight the potential value of promoting sustainable and environmental literacy in supporting healthier dietary patterns. Future studies with more diverse and representative samples are recommended to validate and extend these findings.

Author contributions

GOU: conceptualization, methodology, formal analysis, writing—original draft, writing—review, and editing. SA: data curation, methodology, writing—original draft, writing—review, and editing. AÇ, MA: data curation, writing—original draft, writing—review, and editing. OB, BKB: conceptualization, data curation, formal analysis, writing—review, and editing. The authors have approved the final version submitted.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The research was conducted with "Ethics Committee Approval" from the Erzurum Technical University Ethics Committee (Meeting Number: 03, Decision Number: 17, 2024). The study was carried out following the principles outlined in the Helsinki Declaration. Informed consent was given by each participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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